

PX-200 Centrifugal Pump

Installation, Safety and Troubleshooting Reference

SIMULATED TRAINING DOCUMENT - NOT A MANUFACTURER MANUAL

1. Equipment Identification

Equipment code	PX-200
Equipment type	Horizontal centrifugal process pump
Service	Clean-water circulation training loop
Normal bearing temperature	45-70 C
Temperature warning	Above 80 C
Vibration warning	Above 7.1 mm/s RMS
Manual revision	Training Revision 1.0

System Overview

The PX-200 uses an electric motor, flexible coupling, bearing housing, impeller, mechanical seal and isolation valves. This simulated manual is designed for PulseFix retrieval and citation testing.

SAFETY PREREQUISITES

Stop the pump. Isolate electrical power. Apply lockout/tagout. Close suction and discharge valves. Release stored pressure and verify zero energy. Allow hot surfaces to cool. Wear eye, hand and foot PPE. Never open the casing while it is hot or pressurized.

2. Troubleshooting

High Bearing Temperature with Abnormal Vibration

SAFETY PREREQUISITES

Stop the pump. Isolate electrical power. Apply lockout/tagout. Close suction and discharge valves. Release stored pressure and verify zero energy. Allow hot surfaces to cool. Wear eye, hand and foot PPE. Never open the casing while it is hot or pressurized.

Observed symptoms

Typical reported symptoms are a bearing temperature near 90 C, a rising vibration level, rumbling from the bearing housing, and reduced pump stability.

Possible causes listed in this manual

Documented causes are insufficient or degraded lubrication, bearing wear, shaft or coupling misalignment, loose base bolts, cavitation caused by restricted suction, or operation far from the specified duty point.

Inspection guidance

After full isolation, inspect lubricant level and condition, verify base and coupling bolts, check alignment, rotate the shaft by hand, inspect the suction strainer, and compare vibration at the drive-end and non-drive-end bearings. Do not continue operation when temperature is above 80 C and vibration is abnormal.

Stop / refusal condition

If the safety prerequisites cannot be confirmed, stop. The manual does not authorize diagnosis or repair without isolation, zero-energy verification and required PPE.

3. Authorized Inspection Procedure

SAFETY PREREQUISITES

Stop the pump. Isolate electrical power. Apply lockout/tagout. Close suction and discharge valves. Release stored pressure and verify zero energy. Allow hot surfaces to cool. Wear eye, hand and foot PPE. Never open the casing while it is hot or pressurized.

Step 1. Record temperature and vibration readings before shutdown.

Step 2. Stop the pump using the normal control.

Step 3. Isolate power, lockout/tagout and verify zero energy.

Step 4. Close valves and release pressure.

Step 5. Inspect lubrication, mounting bolts and coupling alignment.

Step 6. Inspect the suction path for blockage or cavitation evidence.

Step 7. Escalate bearing replacement to an authorized technician if roughness or excessive play is found.

Reference Parts

PX200-BRG-DE	Drive-end bearing kit
PX200-BRG-NDE	Non-drive-end bearing kit
PX200-SEAL-01	Mechanical seal kit
PX200-COUP-02	Flexible coupling insert

All work orders generated from this document require explicit supervisor review and approval before persistence or execution.